

English Requirements (Business Rules)

1. Overview of the Business Scenario

The School Management System is a centralized digital platform that supports the academic and administrative operations of a school. It manages data related to students, teachers, parents, classes, subjects, attendance, grades, schedules, communication, and announcements.

The system has four types of users: Admin, Teacher, Student, and Parent, each with specific roles and access permissions.

2. Functional Requirements

2.1 User Authentication & Access Control

- Every user (Admin, Teacher, Student, Parent) must log in using a unique username and password.
- Access is role-based; each user can only perform actions permitted for their role.
- The Admin manages all user accounts (create, edit, deactivate, delete).

2.2 Student Management

- Admin can add, edit, or delete student profiles.
- Teachers can view only the students assigned to their classes.
- Parents can view only their own child's record.

2.3 Teacher Management

- Admin manages teacher profiles.
- Teachers may update only their own basic information.

2.4 Subject & Class Management

- Each Subject has a unique name and description.
- Each Class has:
 - Class Number
 - Floor
 - Capacity (maximum 30 students)
- Students are assigned to exactly one Class and one Academic Year.
- Students may enroll in no more than seven subjects.

2.5 Attendance Management

- Teachers record attendance for each class they teach.
- Attendance can be recorded once per class per day.
- Students and parents can view attendance records.
- Admin can generate attendance reports.

2.6 Grades Management

- Teachers enter and update grades for their subject and assigned classes.
- System automatically calculates grade averages.
- Students and parents can view grades.
- Publishing a grade triggers an instant notification to the student.

2.7 Announcements & Communication

- Admin can post school-wide announcements.
- Teachers can post announcements for their assigned classes.
- Communication rules:
 - Students may message only teachers who teach them.
 - Students cannot message each other.
 - Teachers may message a specific student in their class.
 - Parents communicate only with the Admin (problem reports, report requests).
 - Admin can message parents back.

2.8 Reports

- Admin can print or generate detailed reports for any student or teacher.
- Available reports include performance, attendance, and grades.

2.9 System Security

- System must provide automatic backups.
 - Activity logs track actions of all users.
 - Access is restricted according to user roles.
 - System requires an internet connection (no offline mode).
 - Unique usernames and IDs are required.
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3. Business Rules & Constraints

3.1 User & Role Constraints

- Users cannot access features outside their role.
- Only Admins may delete accounts.
- A student's account may be connected to no more than two parents.

3.2 Student Constraints

- A Student must:
 - Belong to one Class and one Academic Year.
 - Take no more than seven subjects.
 - Have grades recorded in every subject they take.
 - Maintain attendance records for classes they attend.
- Students cannot view other students' grades, attendance, or personal data.

3.3 Teacher Constraints

- A Teacher:
 - Teaches only one subject.
 - Can teach up to three different classes of that subject per academic year.
 - Can only manage content related to classes and subjects they teach.
- Teachers can:
 - Publish announcements to assigned classes.
 - Enter and publish grades (triggering notifications).
 - Message individual students in their classes.

3.4 Parent Constraints

- Parents:
 - May view only their own children's data.
 - Can message Admin for issues or official report requests.
 - Cannot view other students' reports.

3.5 Communication Constraints

- Students may message only teachers who teach them.
 - Students are strictly forbidden from messaging other students.
 - Parents communicate only with Admin.
 - Admin responds to parents and may send reports or notifications.
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4. Examples of Business Rules

- A teacher cannot record attendance for a class not assigned to them.
- A student cannot view grades of other students.
- Only Admins may delete user accounts.
- A parent can view only their own child's academic records.
- A teacher may teach up to three classes but only one subject.
- A student may enroll in maximum seven subjects.